

Mahidol University International College
Computer Engineering International Program

EGCI427 Web Programming
The 1st Trimester, 2026-2027

Instructor:	Dr. Lalita Narupiyakul	E-mail:	Lalita.nar@mahidol.ac.th
Office hours:	Wed. 13:00-14:00	Office:	6253

Course Description

Semistructured data models for the Web (such as JSON, etc.), semistructured data management, associated query languages and systems for Web data, query processing and optimization over semistructured data as well as over multiple distributed sources. Data integration over the Web, warehousing of Web data. Mappings from relational and object-oriented database servers to web data. Some novel data-intensive applications on the Web such as electronic commerce, digital libraries, and distance education as they relate to web database issues will be discussed.

Learning Objectives

- CLO1 Show design of web application and their components.
- CLO2 Implement web application to solve engineering problem.
- CLO3 Deploy new web programming techniques and tools to develop a real-world web application.

Course Information

Date: Wed. and Fri. 10:00-11:50

Room: 6272

Course Material

Microsoft Team: Join Code: **a1v39v6**

Recommended Reading Material:

- Ulises Gascón, "Node.js for Beginners", Packt Publishing. (May, 2024)
- Maya Shavin, "Learning Vue", O'Reilly Media, Inc. (December, 2023)

Note: There is no textbook required for this course. Lecture slides will be provided, and online resources will be recommended.

Measurement and Evaluation of Student Achievement

Item		Date Out	Date Due	Value	AIAS
Assignment 1 (A1) (2 students/group)		30-Oct	20-Nov	20%	Level 4 (50/50 AI)
In-class and Take-home Lab Assignment				20%	Level 4 (50/50 AI)
Group	Course Project and Presentation		04-Dec	30%	Level 4 (50/50 AI)
Lab Quiz			02-Dec	30%	Level 1 (No AI)

Note: (1) Please note that NO late submission will be accepted AFTER due date.

(2) Student's achievement will be evaluated according to the faculty and university standard, using the symbols: A (90%), B+(85%), B(80%), C+(75%), C(70%), D+(65%), D(60%) and F(<60%).

(3) This is not online class. Students must accept any in-class announcements by instructor without online posting.

Assignment Policies

Assignment criteria are specified in the detailed assignment descriptions. Read them carefully to be sure that you have fulfilled all aspects of the requirements. Assignments are DUE ON THE DAY 10:00 AM indicated. No Late Submission will be accepted. There will be 5 marks deduction if you do not follow the assignment instruction.

Extenuating Circumstances

The professor will consider individually, rare extenuating circumstances, which may cause an assignment to be late. Examples of extenuating circumstances include hospitalization, death of a loved one, traffic accidents, etc. The student must provide documentation, such as sick note, to validate the extenuating circumstance. It will be at the professor and student advisor's discretion to work out the extension in this situation.

Academic Dishonesty

Any form of plagiarism is ZERO tolerance. Both parties, who is copying and who is being copied, will get ZERO mark in their assignments.

Course Schedule (Tentative)

Week	Lecture	Topic	Note
1	1	Backend: Env setup, plain Node.js HTTP server	
	2	Backend: ES6+, Promises/async-await, fetch, ES Modules	
2	3	Backend: Introduction to Node.js (1)	
	4	Backend: Introduction to Node.js (2)	
3	5	Backend: Express fundamentals, middleware, routing (1)	
	6	Backend: Express fundamentals, middleware, routing (2)	
4	7	Backend: NOSQL Database Connection and CRUD (1)	
	8	Backend: NOSQL Database Connection and CRUD (2)	
5	9	Backend: Authentication and JWT (1)	
	10	Backend: Authentication and JWT (2)	
6	11	Frontend: Introduction to Vue.js (1)	
	12	Frontend: Introduction to Vue.js (2)	
7	13	Frontend: Vue 3 Composition API, directives, components	
	14	Frontend: Vue Router	A1 Out
8	15	Frontend: Axios integration,	
	16	Full-Stack Framework (1)	
9	17	Full-Stack Framework (2)	
	18	Full-Stack Authentication	
10	19	CI/CD: Docker, Docker Compose, GitHub Actions CI, deployment	
	20	Full-Stack: Build Web Application on Cloud and Web Services (1)	A1 Due
11	21	Full-Stack: Build Web Application on Cloud and Web Services (2)	
	22	Full-Stack: Practice	
12	23	Lab Quiz	03-Dec
	24	Project Presentation	04-Dec

*Please note that this is a tentative schedule and it may be slightly revised based on the progress of classes.

AI Assessment Scale (AIAS)

The AIAS typically consists of five levels, ranging from "No AI" to "Full AI," allowing educators to select the most appropriate level for a given assessment task.

Level	Description	Example
Level 1 (No AI)	Assessments focus on foundational knowledge and skills, where AI assistance is not permitted, ensuring students demonstrate their understanding and abilities without relying on GenAI.	A programming assignment that requires students to write a specific algorithm from scratch without any external help.
Level 2 (Limited AI)	AI tools can be used for specific tasks, such as exploring concepts, suggesting structures, or generating basic code snippets, but students are expected to critically evaluate and refine the AI output.	Students use AI to generate a basic code structure for a program, but they are required to implement the logic and functionality themselves.
Level 3 (Selective AI)	AI can be used in limited and clearly defined ways, such as exploring concepts, suggesting structure, or summarizing key ideas, depending on the assessment's purpose.	Students use AI to find documentation and examples for a specific programming language, but they are required to explain the code and its functionality.
Level 4 (50/50 AI)	Assessments allow for a mix of human and AI input, where students can use AI for certain aspects of the task but still demonstrate their own understanding and critical thinking.	Students use AI to debug a code, but they are required to explain the error and the fix.
Level 5 (Full AI)	Students can use any GenAI tool while completing the assessment, which is appropriate when the focus is on AI-related skills or when the assessment task is designed to encourage creative and innovative use of AI.	Students use AI to generate a complete software program or a complex algorithm, and they are assessed on their ability to use AI effectively and creatively.